🧭 M003 Guide — Counting (x, y, z) · Beginner

Topic: Before 3D · **Course:** 3D Coordinates · **Level:** Beginner · **Time:** about 20 minutes

Pixel art used **two** numbers. A 3D build needs **three**. They always come in the same order:

1st = X	2nd = Y	3rd = Z
right → left	down → up	back → forward
starts at 0	starts at 0	starts at 0

Count x, then y, then z. Always that order.

> 🔴 Stand on your **home spot** (red block) every run. Move your feet, move your build.

1 · Build Your Grid 🎨

Make a striped ruler for each axis, so you can **see** the numbers.

Stand on your home spot. Run this to lay the white floor and the yellow lines:

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
        pos(0, -1, 0),
        pos(15, -1, 15),
        FillOperation.REPLACE)
    blocks.fill(YELLOW_CONCRETE,
        pos(5, -1, 0),
        pos(5, -1, 15),
        FillOperation.REPLACE)
    blocks.fill(YELLOW_CONCRETE,
        pos(10, -1, 0),
        pos(10, -1, 15),
        FillOperation.REPLACE)
    blocks.fill(YELLOW_CONCRETE,
        pos(0, -1, 5),
        pos(15, -1, 5),
        FillOperation.REPLACE)
    blocks.fill(YELLOW_CONCRETE,
        pos(0, -1, 10),
        pos(15, -1, 10),
        FillOperation.REPLACE)
player.on_chat("grid", on_run)
```

Now place the three rulers by hand — **yellow, black, yellow, black** all the way. All three start on the block under your feet:

- **X ruler** — one block at a time, **to your left**.
- **Z ruler** — one block at a time, **forward**.
- **Y ruler** — a pillar straight **up**.

> 🟧 ⬛ The stripes are there so you can count without guessing.

2 · Start at the Corner 🔴

The three rulers all meet on **one block**. That block is your home spot.

Its coordinate is **(0, 0, 0)**. Every count starts there.

From the corner	You get
stand still	(0, 0, 0)
1 step **left**	(1, 0, 0)
1 block **up**	(0, 1, 0)
1 step **forward**	(0, 0, 1)

x, y and z all start at which number? Circle one: 0 · 1

Where is (0, 0, 0)? Circle one: the corner · one block up

3 · The Three Rulers 📏

Each ruler starts at the corner and counts **0, 1, 2, 3 ...**

X — right → left

Y — down → up

Z — back → forward

```
place red block at (1, 0, 0)   # 1 step left
place red block at (0, 1, 0)   # 1 block up
place red block at (0, 0, 1)   # 1 step forward
```

Which ruler goes to your left? Circle one: x · y · z

A block at y = 3 is ...? Circle one: on the ground · 3 blocks up

Which ruler goes forward? Circle one: x · y · z

4 · Put Them Together 🧭

Read **(5, 2, 3)** in three steps:

Step	Number	Ruler	Do this
1	x = 5	right → left	go 5 to your left
2	y = 2	down → up	go 2 up (the ground is 0)
3	z = 3	back → forward	go 3 forward

Which coordinate is on the ground? Circle one: (2, 1, 2) · (2, 0, 2)

These two blocks are stacked. Which number grew?

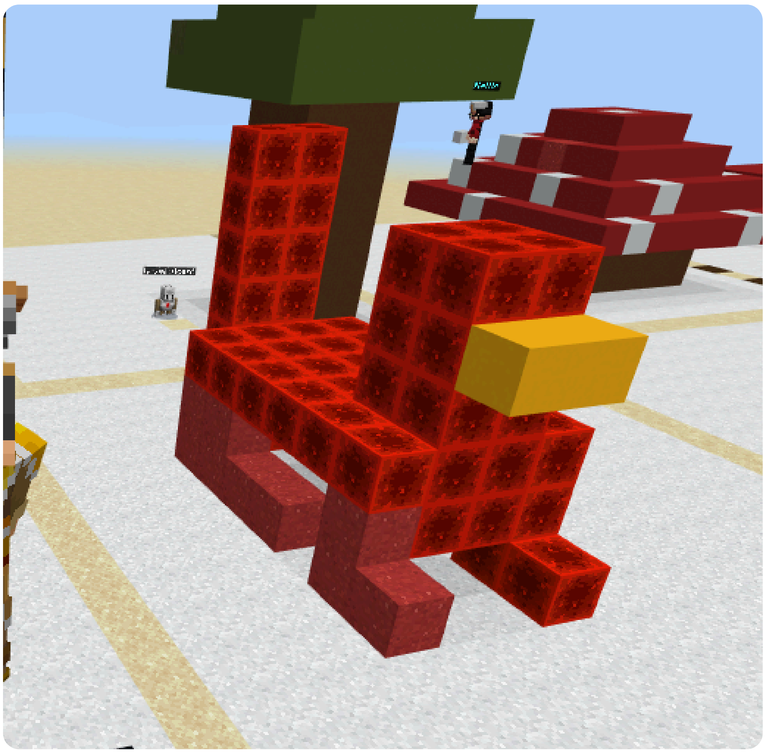
```
place red block at (7, 1, 2)
place red block at (7, 2, 2)
```

Circle one: x · y · z

5 · Show Your Work 📸 🎥

Warm-up: stand on your home spot and place one block at **(3, 0, 3)**. Check it against your rulers.

Build the chicken. Count x, then y, then z for every block.



> 🧱 Recipe (red nether brick, yellow concrete for the beak):
>
> - legs: (4, 0, 4) and (4, 0, 5)
> - body: (3, 1, 4) to (6, 2, 5)
> - tail: (2, 3, 4) and (2, 3, 5)
> - head: (6, 3, 4) to (7, 4, 5)
> - beak: yellow at (8, 3, 4) and (8, 3, 5)

If you finish early, give it a comb on top of the head.

Record **one video** (a phone is fine). Show two things:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Count one block out loud — x, then y, then z.

Say these out loud, filling in the blanks:

> Today I built _____.
>
> I built it using this code: _____.

> In this code I used ______.
>
> The hardest part was ______.
>
> That part was hard because ______.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.